MATHEMATICS · CLASS 9/10 · SPEED & MEMORY

Mental Math Toolkit

Squares, cubes, tables and the shortcuts that turn a 2-minute calculation into a 10-second one. Memorise these once, and every chapter gets faster — from Heron's formula to quadratic equations.

PART 1

Squares & Cubes to Memorise

The single highest-value list in this whole book — used in Heron's formula, quadratic equations, surds, and every "spot the perfect square" question.

SQUARES, 1–30

1–10: $1^2=1$ · $2^2=4$ · $3^2=9$ · $4^2=16$ · $5^2=25$ · $6^2=36$ · $7^2=49$ · $8^2=64$ · $9^2=81$ · $10^2=100$

11–20: $11^2=121$ · $12^2=144$ · $13^2=169$ · $14^2=196$ · $15^2=225$ · $16^2=256$ · $17^2=289$ · $18^2=324$ · $19^2=361$ · $20^2=400$

21–30: $21^2=441$ · $22^2=484$ · $23^2=529$ · $24^2=576$ · $25^2=625$ · $26^2=676$ · $27^2=729$ · $28^2=784$ · $29^2=841$ · $30^2=900$

CUBES, 1–20

1–10: $1^3=1$ · $2^3=8$ · $3^3=27$ · $4^3=64$ · $5^3=125$ · $6^3=216$ · $7^3=343$ · $8^3=512$ · $9^3=729$ · $10^3=1000$

11–20: $11^3=1331$ · $12^3=1728$ · $13^3=2197$ · $14^3=2744$ · $15^3=3375$ · $16^3=4096$ · $17^3=4913$ · $18^3=5832$ · $19^3=6859$ · $20^3=8000$

SQUARE ROOTS & CUBE ROOTS WORTH KNOWING

Perfect squares/cubes e.g. $\sqrt{625} = 25$, since $25^2 = 625$.	read the tables above in reverse
$\sqrt{2}$	≈ 1.414
$\sqrt{3}$	≈ 1.732
$\sqrt{5}$	≈ 2.236
$\sqrt{6}$	≈ 2.449
$\sqrt{7}$	≈ 2.646
$\sqrt{8}$	≈ 2.828
$\sqrt{10}$	

	≈ 3.162
∛2	≈ 1.260
∛3	≈ 1.442
∛5	≈ 1.710

PART 2

Multiplication Tables, 11–20

Past the usual 1–10, these are what actually save time in exams — especially for factorising and mental multiplication.

- **×11:** 11, 22, 33, 44, 55, 66, 77, 88, 99, 110
- **×12:** 12, 24, 36, 48, 60, 72, 84, 96, 108, 120
- **×13:** 13, 26, 39, 52, 65, 78, 91, 104, 117, 130
- **×14:** 14, 28, 42, 56, 70, 84, 98, 112, 126, 140
- **×15:** 15, 30, 45, 60, 75, 90, 105, 120, 135, 150
- **×16:** 16, 32, 48, 64, 80, 96, 112, 128, 144, 160
- **×17:** 17, 34, 51, 68, 85, 102, 119, 136, 153, 170
- **×18:** 18, 36, 54, 72, 90, 108, 126, 144, 162, 180
- **×19:** 19, 38, 57, 76, 95, 114, 133, 152, 171, 190
- **×20:** 20, 40, 60, 80, 100, 120, 140, 160, 180, 200

PART 3

Fractions ↔ Decimals ↔ Percentages

Recognise these on sight and you'll never need to long-divide them again.

- **Halves & quarters:** 1/2=0.5=50% · 1/4=0.25=25% · 3/4=0.75=75%
- **Thirds & sixths:** 1/3≈0.333=33.33% · 2/3≈0.667=66.67% · 1/6≈0.167=16.67% · 5/6≈0.833=83.33%
- **Fifths:** 1/5=0.2=20% · 2/5=0.4=40% · 3/5=0.6=60% · 4/5=0.8=80%

Eighths: 1/8=0.125=12.5% · 3/8=0.375=37.5% · 5/8=0.625=62.5% · 7/8=0.875=87.5%

Others: 1/9≈0.111=11.11% · 1/10=0.1=10% · 1/16=0.0625=6.25% · 1/20=0.05=5% · 1/25=0.04=4%

PART 4

Divisibility Rules

Spot a factor instantly — no division required.

By 2	**last digit is even (0,2,4,6,8)**
By 3 e.g. 741 → 7+4+1=12, divisible by 3.	**digit sum divisible by 3**
By 4 e.g. 3,116 → 16 is divisible by 4.	**last two digits divisible by 4**
By 5	**last digit is 0 or 5**
By 6	**divisible by both 2 AND 3**
By 8	**last three digits divisible by 8**
By 9 e.g. 4,527 → 4+5+2+7=18, divisible by 9.	**digit sum divisible by 9**
By 10	**last digit is 0**
By 11 **(sum of digits at odd places) − (sum at even places) is 0 or divisible by 11** e.g. 2,728 → (2+2)-(7+8) = 4-15 = -11 → divisible by 11.	

PART 5

Quick Calculation Tricks

Mental-math shortcuts that turn multi-step arithmetic into one or two moves — useful whenever a calculator isn't allowed.

MULTIPLY BY 5

Multiply by 10, then halve it.

48 × 5 → 48 × 10 = 480 → 480 ÷ 2 = **240**

MULTIPLY BY 25

Multiply by 100, then divide by 4.

36 × 25 → 36 × 100 = 3600 → 3600 ÷ 4 = **900**

MULTIPLY BY 9

Multiply by 10, then subtract the number once.

23 × 9 → 230 - 23 = **207**

MULTIPLY A 2-DIGIT NUMBER BY 11

Add the two digits and slot the sum in the middle (carry over if the sum is 10 or more).

34 × 11 → 3 _ 4, and 3+4=7 → **374**. For 57 × 11: 5+7=12, carry the 1 → 5(+1) _ 2 → **627**

SQUARE A NUMBER ENDING IN 5

For a number "a5", the answer is a × (a+1), followed by 25.

35^2 → a=3 → 3×4=12 → **1225**. 85^2 → a=8 → 8×9=72 → **7225**

SQUARE A NUMBER NEAR A ROUND BASE

Use $(a-b)^2 = a^2-2ab+b^2$ with the nearest hundred as a.

$98^2 = (100-2)^2$ = 10000 - 400 + 4 = **9604**

MULTIPLY TWO NUMBERS NEAR THE SAME BASE

Find each number's difference from the base (here, 100); cross-subtract for the first part, multiply the differences for the last part.

97 × 96 (differences: -3, -4) → 97-4 = 93; 3×4=12 → **9312**

DIVIDE BY 5

Multiply by 2, then divide by 10 (i.e. move the decimal point).

64 ÷ 5 → 64 × 2 = 128 → 128 ÷ 10 = **12.8**

DIVIDE BY 25

Multiply by 4, then divide by 100.

300 ÷ 25 → 300 × 4 = 1200 → 1200 ÷ 100 = **12**

FIND A PERCENTAGE QUICKLY

10% = shift the decimal one place left. 5% = half of 10%. 1% = shift two places left. Add these building blocks together for any percentage.

15% of 340 → 10% = 34, 5% = 17 → 34+17 = **51**

MULTIPLY BY 11 — ANY SIZE NUMBER

Multiply by 10, then add the number once more: N × 11 = N×10 + N.

123 × 11 → 1230 + 123 = **1353**

MULTIPLY BY 4

Double it, then double it again.

37 × 4 → 37×2=74 → 74×2 = **148**

MULTIPLY BY 8

Double it three times.

12 × 8 → 24 → 48 → **96**

MULTIPLY BY 6

Multiply by 5, then add the number once more.

23 × 6 → 23×5=115 → 115+23 = **138**

MULTIPLY BY 12

Multiply by 10, then add double the number (×12 = ×10 + ×2).

23 × 12 → 230 + 46 = **276**

MULTIPLY BY 15

Multiply by 10, then add half of that result.

24 × 15 → 24×10=240 → half = 120 → 240+120 = **360**

MULTIPLY BY 99 (OR 999)

Multiply by 100 (or 1000), then subtract the number once.

47 × 99 → 4700 - 47 = **4653**

SQUARE A NUMBER NEAR 50

Use $(50\pm a)^2 = 2500 \pm 100a + a^2$.

53^2 → a=3 → 2500+300+9 = **2809**. 47^2 → a=3 → 2500-300+9 = **2209**

SUM OF CONSECUTIVE NUMBERS (GAUSS'S TRICK)

1 + 2 + … + n = n(n+1)/2 — no need to add term by term.

1+2+…+50 → 50×51/2 = **1275**

BONUS

Powers of 2 & Primes up to 100

Not always exam-critical, but they turn up often enough to be worth having ready.

Powers of 2: $2^1=2 \cdot 2^2=4 \cdot 2^3=8 \cdot 2^4=16 \cdot 2^5=32 \cdot 2^6=64 \cdot 2^7=128 \cdot 2^8=256 \cdot 2^9=512 \cdot 2^{10}=1024$

Primes up to 100 (25 of them): 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97

Ten minutes a day reciting these out loud beats an hour of silent staring the night before an exam.

Fast hands, calm mind.